Elenchus

Generating Knowledge Bases from Prover-Skeptic Dialogues

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The implementation problem in knowledge engineering

> *"I told you that I started some AI companies – the reason I am not wealthy right now – the reason that I don't have tens of millions of dollars. . . is that bottleneck – it was too hard to get knowledge into the knowledge bases."*
>
> — *Edward Feigenbaum, quoted in* Augier *et al. (2002)*

Knowledge engineering (KE): the collection of activities for formalizing knowledge for use in information systems (Studer et al., 1998)

The implementation problem: The difficulty of the formalization (Dutilh Novaes, 2012) of expert knowledge expressed in natural language

Seventy years of attempts to resolve the implementation problem

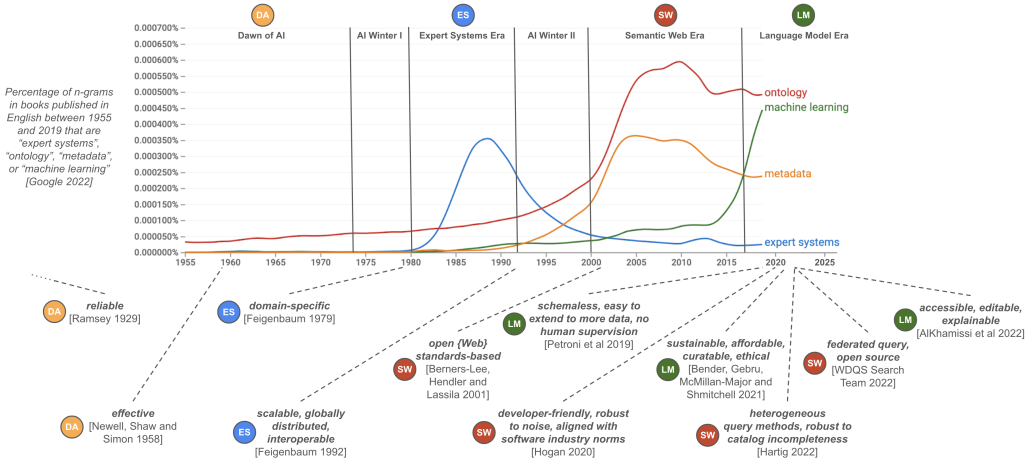

Figure from Allen (2023)

An inferentialist reconceptualization of knowledge engineering?

- We argue this problem stems from an **internalist conception of knowledge engineering** (Forsythe, 1993)
- Drawing on inferentialist semantics (Brandom, 1994; Hlobil et al., 2025), we propose an alternative view of knowledge engineering as **collaborative meaning-making**: a negotiation of which inferential relationships hold among statements
- Could collaborative meaning-making through **dialogue in natural language** address the implementation problem by eliminating the need for expertise in formalization?
- Furthermore: **could large language models (LLMs) play a role in such dialogues?**

Evidence that LLMs could play a role in an inferentialist practice of KE

- **Generating material inferences:** *Classifiers-as-intensions* can be built using zero-shot prompting of an LLM using natural language intensional definitions of concepts and relations (Allen, 2025a)
- **Articulating reasons:** *Rationales* produced by classifiers-as-intensions can guide resolution of factual or metalinguistic disagreements between experts, knowledge graphs and LLMs (Allen, 2025a; Allen et al., 2025a,b)
- **Checking coherence of bilateral positions:** *LLM-grounded evaluation* of commitments and denials enables consistency checking that respects the non-classical structure of incomplete or conflicting positions (Allen et al., 2025c)
- **Epistemic transparency:** *Bilateral generalized truth values* provide actionable insights into the doxastic state of an LLM (Allen et al., 2025c)

Elenchus: a bilateral dialectical protocol for knowledge acquisition

- An instance of a **Scenario 2 prover-skeptic dialogue** (Dutilh Novaes, 2020):
  - A human **respondent** develops a position through dialogue with an LLM **opponent**
  - The respondent proposes **commitments** and **denials**
  - The opponent identifies **tensions**—claims that these are jointly incoherent
  - The respondent resolves tensions by **retraction**, **refinement**, or **contestation**
- The LLM serves as a **defeasible derivability oracle**: its identified tensions are not authoritative but subject to negotiation
- Implemented (Allen, 2025b) as **a Claude Code agent** (Anthropic, 2025) that stores dialectical state as GitHub issues (GitHub, 2025)
- Inspired by discussions with Thomas Ferguson (RPI), Shay Logan (KSU), Marcus Rossberg (UConn), and Teresa Kouri Kessel (Old Dominion)

The Elenchus agent loop

```
Load dialectical state ⟨[Γ : Δ], T, I⟩ from GitHub
while session active do
    Wait for respondent's speech act        ▷ commitment, denial, tension response
    if commitment (+P) or denial (−P) then
        Update position; check coherence
        if tension detected then add to T as sequent X ⊢ Y    ▷ X ⊆ Γ, Y ⊆ Δ
        end if
    else if resolves tension X ⊢ Y ∈ T then
        if retraction or refinement then update [Γ : Δ]; move X ⊢ Y to I
        else if contestation then remove X ⊢ Y from T
        end if
    end if
    Record state to GitHub
    if should probe then raise Socratic challenge
    end if
    Display open tensions T and challenges
end while
```

Mapping Elenchus dialectical states to NMMS material bases

- Elenchus dialectical states map directly to **material bases** in the sense of Hlobil and Brandom's NMMS sequent calculus
- The resulting structure satisfies **Containment**, enabling NMMS to elaborate logical vocabulary that makes explicit the inferential relationships negotiated in dialogue
- This provides the formal representation—propositions plus their inferential structure—that constitutes a **knowledge base**

Question for discussion: Does this mapping make sense?

Mapping NMMS logical vocabularies to knowledge bases?

- Every material implication traces to a specific dialogue move: a tension proposed by the LLM and accepted by the respondent
- Unlike knowledge bases extracted from corpora, where justifications may be opaque, Elenchus produces a **complete, auditable record** of the knowledge acquisition process
- However, knowledge bases typically involve **universally- or existentially-quantified commitments** (e.g., "all birds fly")
- Tension-checking and export to classical reasoners could exploit the standard **reduction to propositional logic** over finite domains to support this

Question for discussion: Is there a path to a first-order NMMS that would eliminate the need for this reduction?

References I

- Allen, Bradley P (2023). "A Brief History of Knowledge Engineering: A Practitioner's Perspective". In: *Dagstuhl Reports* 12.9, pp. 65–72.
- — (2025a). "Conceptual Engineering Using Large Language Models". In: *Philosophy of Artificial Intelligence: The State of the Art)*. Ed. by Guido Löhr Vincent C. Müller Leonard Dung and Aliya Rumana. To appear. Springer Nature. url: https://doi.org/10.48550/arXiv.2404.03732.
- — (2025b). *Elenchus*. GitHub repository. url: https://github.com/bradleypallen/elenchus.
- Allen, Bradley P. and Paul Groth (2025a). "Evaluating Class Membership Relations in Knowledge Graphs using Large Language Models". In: *The Semantic Web: ESWC 2024 Satellite Events*. Ed. by Albert Meroño-Peñuela et al. Vol. 15344. Lecture Notes in Computer Science. Cham: Springer. isbn: 978-3-031-78951-9. doi: 10.1007/978-3-031-78952-6_2. url: https://doi.org/10.1007/978-3-031-78952-6_2.
- Allen, Bradley P. and Paul T. Groth (2025b). "A Benchmark for the Detection of Metalinguistic Disagreements between LLMs and Knowledge Graphs". In: *Proceedings of the Special Session on Harmonising Generative AI and Semantic Web Technologies (HGAIS 2024) co-located with the 23rd International Semantic Web Conference (ISWC 2024)*. Vol. 3953. CEUR Workshop Proceedings. Baltimore, Maryland: CEUR-WS.org. url: https://ceur-ws.org/Vol-3953/.

References II

Allen, Bradley P. et al. (2025c). "Sound and Complete Neurosymbolic Reasoning with LLM-Grounded Interpretations". In: *Proceedings of The 19th International Conference on Neurosymbolic Learning and Reasoning*. Ed. by Leilani H. Gilpin et al. Vol. 284. Proceedings of Machine Learning Research. PMLR, pp. 392–419. url: https://proceedings.mlr.press/v284/allen25a.html.

Anthropic (2025). *Claude Code Overview*. URL accessed: 2025-10-14. url: https://docs.claude.com/en/docs/claude-code/overview.

Augier, Mie-Sophia and Morten Thanning Vendelø (2002). *An Interview with Edward A. Feigenbaum*. Interview. Copenhagen, Denmark: Copenhagen Business School, Department of Informatics. url: https://www.researchgate.net/publication/236904576_An_Interview_with_Edward_A_Feigenbaum.

Brandom, Robert (1994). *Making it explicit: Reasoning, representing, and discursive commitment*. Harvard University Press.

Dutilh Novaes, Catarina (2012). *Formal languages in logic: A philosophical and cognitive analysis*. Cambridge University Press.

— (2020). *The dialogical roots of deduction: Historical, cognitive, and philosophical perspectives on reasoning*. Cambridge University Press. doi: 10.1017/9781108800792.

References III

- Forsythe, Diana E (1993). "Engineering knowledge: The construction of knowledge in artificial intelligence". In: *Social studies of science* 23.3, pp. 445–477.
- GitHub (2025). *About issues*. URL accessed: 2025-02-01. url: https://docs.github.com/en/issues/tracking-your-work-with-issues/about-issues.
- Hlobil, Ulf and Robert B Brandom (2025). *Reasons for logic, logic for reasons: Pragmatics, semantics, and conceptual roles*. Routledge.
- Studer, Rudi, V Richard Benjamins, and Dieter Fensel (1998). "Knowledge engineering: Principles and methods". In: *Data & knowledge engineering* 25.1-2, pp. 161–197. doi: 10.1016/S0169-023X(97)00056-6.